Sexual Reproduction in Humans

Exam code: 0610 & 0970

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The male reproductive system

Structures of the male reproductive system

- The male reproductive system has several key components
- The structure and function of these components can be seen in the table below

Structure	Function
Prostate gland	Produces a fluid that mixes with sperm cells to form semen
Sperm duct	Sperm passes through the sperm duct to be mixed with fluids produced by the glands before being passed into the urethra for ejaculation
Urethra	Tube running down the centre of the penis that can carry out urine or semen, a ring of muscle

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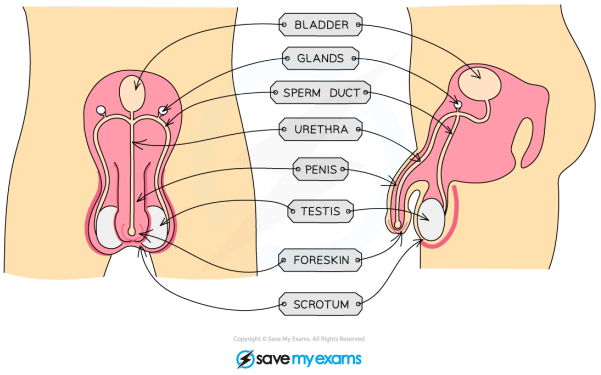
Got it

Help me understand

Testis	Contained within a bag of skin (scrotum) and produces sperm (male gamete) and testosterone hormone
Scrotum	Sac supporting the testes outside the body to ensure that sperm are kept at a temperature slightly lower than body temperature
Penis	Passes urine out of the body from the bladder and allows semen to pass into the vagina of a woman during sexual intercourse

The male reproductive system diagram



The male reproductive system

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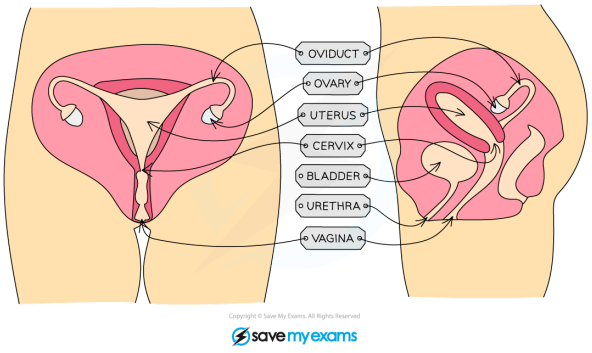
The female reproductive system

Structures of the female reproductive system

seen in the table below

Structure	Function
Oviduct	Connects the ovary to the uterus and is lined with ciliated cells to push the released ovum down it. Fertilisation occurs here
Ovary	Contains ova (female gametes) which will mature and develop when hormones are released
Uterus	Muscular bag with a soft lining where the fertilised egg (zygote) will be implanted to develop into a foetus
Cervix	Ring of muscle at the lower end of the uterus to keep the developing foetus in place during pregnancy
Vagina	The muscular tube that leads to the inside of the woman's body where the males penis will enter during sexual intercourse and sperm are deposited

The female reproductive system diagram



The female reproductive system

Gametes & Fertilisation

What is fertilisation?

- Fertilisation is the **fusion of the nuclei from a male gamete (sperm cell) and a female gamete (egg cell)**
- It occurs in the **oviducts**
- Gametes have **adaptations** to increase the chances of fertilisation and successful development of an embryo

Fertilisation diagram

The sperm enters the egg cell during fertilisation which usually occurs in the oviduct

Adaptations of Gametes

Human gametes

- The human gametes are the **egg** and the **sperm** cells

Egg and sperm cell diagram

Comparing sperm and egg cells

Adaptive features of the gametes

- These **adaptive features** are compared in the table below

Gamete	Adaptive feature	Function
Sperm	has a flagellum (tail)	allows the sperm to swim towards the egg
	contains **enzymes** in the head region (acrosome)	to digest a route into the egg for fertilisation
	contains many mitochondria	to provide energy for movement of the flagellum
Egg	cytoplasm contains a store of energy	to provide energy for cell division in the developing zygote after fertilisation
	jelly-like coating that changes after fertilisation	to make an impenetrable barrier after fertilisation to prevent more sperm entering the egg

Comparison of Male & Female Gametes

Comparative Feature	Sperm	Egg
Size	Very small (45 µm)	Large (0.15 mm)
Structure	Head region, flagellum, many structural adaptations	Round cell with few structure adaptations, covered in a jelly coating
Motility	Capable of locomotion	Not capable of locomotion
Numbers	Produced every day in huge numbers (around 100 million per day)	Thousands of immature eggs in each ovary, but only one is released each month

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Pregnancy: growth & development of the fetus

- After fertilisation in the **oviduct**, the zygote travels towards the **uterus**
  - This takes about 3 days, during which time the zygote will divide several times to form a ball of cells known as an **embryo**
- In the uterus, the embryo embeds itself in the thick lining (**implantation**) and continues to grow and develop
- The gestation period for humans is **9 months**
  - Major **development of organs** takes place within the **first 12 weeks**, during which time the embryo gets nutrients from the mother by **diffusion through the uterus lining**
  - After this point the organs are all in place, the **placenta** has formed and the embryo is now called a **fetus**
  - The remaining gestation time is used by the fetus to **grow bigger in size**
- The fetus is surrounded by an **amniotic sac** which contains **amniotic fluid** (made from the mother's blood plasma)
  - This protects the fetus during development by **cushioning it from bumps** to the mother's abdomen
- The **umbilical cord** joins the fetus's blood supply to the **placenta** for exchange of nutrients and removal of waste products

The fetus in the uterus

The placenta & umbilical cord: extended

What is the the placenta and the umbilical cord?

- During the gestation period the fetus develops and grows by gaining the **glucose, amino acids, fats**, **water** and **oxygen** it needs from the mother's blood
- The bloods run opposite each other, never mixing, in the **placenta**
- The fetus's blood connects to and from the placenta by the **umbilical cord**
- The mother's blood also absorbs the **waste** from the fetus's blood in the placenta; substances like **carbon dioxide** and **urea** are removed from the fetus's blood so that they do not build up to dangerous levels
- Movement of all molecules across the placenta occurs by **diffusion** due to **difference in concentration gradients**
- The placenta is adapted for this diffusion by having a l**arge surface area** and a **thin wall** for efficient diffusion

Toxins and pathogens

- The placenta acts as a **barrier** to prevent toxins and pathogens getting into the fetus's blood
- Not all **toxin molecules** or **pathogenic organisms** (such as **viruses**, eg **rubella**) are stopped from passing through the placenta (this usually depends on the size of the molecule)
- This is why pregnant women are advised **not to smoke during pregnancy** as molecules like **nicotine** can pass across the placenta

Placenta and umbilical cord diagram

The placenta

Examiner Tips and Tricks

It is worth learning at least two specific substances that move in either direction across the placenta – this is a common exam question and non-specific answers such as 'waste products' and 'nutrients' will not get any marks!

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